

BSDS V: Regression Techniques
Problems Set 2: Least Squares Regression
Detailed Step-by-Step Solution Set

Prepared directly from the uploaded ISI lecture notes and PS2

How to use this solution set

This solution set is deliberately written as a *derivation document*. For every problem the order is:

1. the complete question from the assignment;
2. the exact pieces of theory from the uploaded lecture notes that are required;
3. a sentence explaining what the theory is doing in this problem;
4. the derivation, with every algebraic transition explained;
5. the final answer boxed at the end.

Notation used throughout. The assignment writes

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n.$$

The design matrix has one intercept column and p regressor columns, hence it has $p + 1$ columns. When $\mathbf{X}$ has full column rank, $\text{rank}(\mathbf{X}) = p + 1$ and $(\mathbf{X}^\top \mathbf{X})^{-1}$ exists.

Source note. The problem statements below are transcribed from the uploaded PS2, and the theory is taken from the uploaded “Regression Techniques” lecture notes. The assignment’s Q7 contains an apparent OCR typo; the meaningful statement is explained explicitly in that solution rather than silently changing it.

Part I

A: Problems without any distributional assumption on errors

1 A1. Centered covariates: multiple OLS versus separate simple regressions

Question (as assigned)

1. Consider the least square solution of $\hat{\beta}_{OLS}$, obtained from the linear regression model in (1), where the covariates are centered, i.e., $\mathbf{1}^\top \mathbf{x}^{(j)} = 0$ for all $j = 1, \dots, p$. Further, consider the j -th simple linear regression model

$$Y_i = \beta_0 + \beta_j x_{i,j} + \varepsilon_i, \quad i = 1, \dots, n,$$

and let $\hat{\beta}_{j,S}$ be the solution to the simple linear regression model, and

$$\hat{\beta}_S = [\bar{Y} \quad \hat{\beta}_{1,S} \quad \dots \quad \hat{\beta}_{p,S}]^\top.$$

- (a) Compare $\hat{\beta}_{OLS}$ and $\hat{\beta}_S$.
- (b) When will these two solutions be equal?

Required theory from the lecture notes

The notes define the linear model and explain the interpretation of multiple-regression coefficients. In a multiple regression model,

$$Y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i,$$

β_j measures the change in the response associated with a one-unit change in x_j *while the other covariates are kept fixed*. In separate simple regressions, the coefficient attached to x_j describes the relationship of Y with x_j without controlling for the other regressors. The notes explicitly state that these interpretations need not agree, and even their signs can reverse (Simpson's paradox). The least-squares estimator satisfies

$$\hat{\beta}_{OLS} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

when $\mathbf{X}^\top \mathbf{X}$ is invertible.¹

What centering changes

Let

$$\mathbf{X}_c = [\mathbf{1} : \mathbf{x}^{(1)} : \dots : \mathbf{x}^{(p)}]$$

with $\mathbf{1}^\top \mathbf{x}^{(j)} = 0$. Then

$$\mathbf{X}_c^\top \mathbf{X}_c = \begin{bmatrix} \mathbf{1}^\top \mathbf{1} & \mathbf{1}^\top \mathbf{x}^{(1)} & \dots & \mathbf{1}^\top \mathbf{x}^{(p)} \\ (\mathbf{x}^{(1)})^\top \mathbf{1} & (\mathbf{x}^{(1)})^\top \mathbf{x}^{(1)} & \dots & (\mathbf{x}^{(1)})^\top \mathbf{x}^{(p)} \\ \vdots & \vdots & \ddots & \vdots \\ (\mathbf{x}^{(p)})^\top \mathbf{1} & (\mathbf{x}^{(p)})^\top \mathbf{x}^{(1)} & \dots & (\mathbf{x}^{(p)})^\top \mathbf{x}^{(p)} \end{bmatrix} = \begin{bmatrix} n & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{Z}^\top \mathbf{Z} \end{bmatrix},$$

¹This is the least-squares normal-equation result in Section 2.1 of the supplied notes. The notes also explain the geometric interpretation as projection onto $\mathcal{C}(\mathbf{X})$.

where $\mathbf{Z} = [\mathbf{x}^{(1)} : \dots : \mathbf{x}^{(p)}]$.

Thus the intercept column is orthogonal to every centered regressor.

Solution to (a)

For the centered design,

$$\hat{\beta}_{OLS} = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top \mathbf{Y}.$$

Because the matrix is block diagonal,

$$(\mathbf{X}_c^\top \mathbf{X}_c)^{-1} = \begin{bmatrix} 1/n & \mathbf{0}^\top \\ \mathbf{0} & (\mathbf{Z}^\top \mathbf{Z})^{-1} \end{bmatrix}.$$

Also,

$$\mathbf{X}_c^\top \mathbf{Y} = \begin{bmatrix} \mathbf{1}^\top \mathbf{Y} \\ \mathbf{Z}^\top \mathbf{Y} \end{bmatrix} = \begin{bmatrix} n\bar{Y} \\ \mathbf{Z}^\top \mathbf{Y} \end{bmatrix}.$$

Therefore,

$$\hat{\beta}_{0,OLS} = \frac{1}{n} \mathbf{1}^\top \mathbf{Y} = \bar{Y},$$

and the slope vector is

$$\hat{\beta}_{1:p,OLS} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{Y}.$$

Now consider the j -th simple regression. Its slope is

$$\hat{\beta}_{j,S} = \frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j)(Y_i - \bar{Y})}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}.$$

Since the regressor is centered, $\bar{x}_j = 0$, so

$$\hat{\beta}_{j,S} = \frac{\sum_i x_{ij}(Y_i - \bar{Y})}{\sum_i x_{ij}^2}.$$

Because $\sum_i x_{ij} = 0$,

$$\sum_i x_{ij} \bar{Y} = \bar{Y} \sum_i x_{ij} = 0,$$

so

$$\hat{\beta}_{j,S} = \frac{\sum_i x_{ij} Y_i}{\sum_i x_{ij}^2}.$$

Thus, in general,

$$\boxed{\hat{\beta}_{j,S} = \frac{\mathbf{x}_j^\top \mathbf{Y}}{\mathbf{x}_j^\top \mathbf{x}_j}}$$

whereas the multiple-regression coefficient is the j -th component of

$$(\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{Y}.$$

Therefore, centering by itself does not make the separate simple-regression slopes equal to the multiple-regression slopes.

Solution to (b): when are they equal?

The two estimates are equal precisely when each regressor is orthogonal to all the other regressors. Indeed, if

$$\mathbf{x}_k^\top \mathbf{x}_j = 0 \quad (k \neq j),$$

then

$$\mathbf{Z}^\top \mathbf{Z} = \text{diag}(\|\mathbf{x}_1\|^2, \dots, \|\mathbf{x}_p\|^2),$$

so

$$\hat{\beta}_{j,OLS} = \frac{\mathbf{x}_j^\top \mathbf{Y}}{\mathbf{x}_j^\top \mathbf{x}_j} = \hat{\beta}_{j,S}.$$

The intercept is already equal because the covariates are centered:

$$\hat{\beta}_{0,OLS} = \bar{Y} = \hat{\beta}_{0,S}.$$

$\hat{\beta}_{OLS} = \hat{\beta}_S$ when the centered regressors are pairwise orthogonal.

Interpretation. The simple-regression coefficient ignores correlations with the other regressors. When the regressors are orthogonal, there is nothing to control for: projecting onto one regressor direction does not interfere with the other regressor directions.

2 A2. Completely orthogonal design

Question (as assigned)

2. Consider the regression model in (1), and let the design matrix satisfy the following orthogonality property:

$$\mathbf{1}^\top \mathbf{x}^{(j)} = 0, \quad (\mathbf{x}^{(k)})^\top \mathbf{x}^{(j)} = 0, \quad k \neq j, \quad j, k = 1, \dots, p.$$

- (a) Find the expression of $\hat{\beta}_{OLS}$ under this situation.
- (b) Find the expression of the residual sum of squares (RSS) under this situation.

Required theory from the lecture notes

Least squares minimizes

$$S(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|^2 = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta).$$

The normal equations are

$$\mathbf{X}^\top \mathbf{X} \hat{\beta}_{OLS} = \mathbf{X}^\top \mathbf{Y},$$

and for full column rank,

$$\hat{\beta}_{OLS} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

The residual vector is

$$\mathbf{e} = \mathbf{Y} - \mathbf{X} \hat{\beta}_{OLS} = (\mathbf{I} - P_X) \mathbf{Y}, \quad P_X = \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top,$$

and

$$\text{RSS} = \|\mathbf{e}\|^2 = \mathbf{Y}^\top (\mathbf{I} - P_X) \mathbf{Y}.$$

Solution to (a)

Under the stated assumptions,

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} n & 0 & \cdots & 0 \\ 0 & \|\mathbf{x}^{(1)}\|^2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \|\mathbf{x}^{(p)}\|^2 \end{bmatrix}.$$

Hence

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \begin{bmatrix} 1/n & 0 & \cdots & 0 \\ 0 & 1/\|\mathbf{x}^{(1)}\|^2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\|\mathbf{x}^{(p)}\|^2 \end{bmatrix}.$$

Also,

$$\mathbf{X}^\top \mathbf{Y} = \begin{bmatrix} \mathbf{1}^\top \mathbf{Y} \\ (\mathbf{x}^{(1)})^\top \mathbf{Y} \\ \vdots \\ (\mathbf{x}^{(p)})^\top \mathbf{Y} \end{bmatrix}.$$

Therefore,

$$\hat{\beta}_0 = \bar{Y}, \quad \hat{\beta}_j = \frac{(\mathbf{x}^{(j)})^\top \mathbf{Y}}{\|\mathbf{x}^{(j)}\|^2}, \quad j = 1, \dots, p.$$

Solution to (b): RSS

The fitted vector is

$$\hat{\mathbf{Y}} = \hat{\beta}_0 \mathbf{1} + \sum_{j=1}^p \hat{\beta}_j \mathbf{x}^{(j)}.$$

Because the basis vectors $\mathbf{1}, \mathbf{x}^{(1)}, \dots, \mathbf{x}^{(p)}$ are mutually orthogonal, Pythagoras gives

$$\|\hat{\mathbf{Y}}\|^2 = \hat{\beta}_0^2 \|\mathbf{1}\|^2 + \sum_{j=1}^p \hat{\beta}_j^2 \|\mathbf{x}^{(j)}\|^2.$$

Since least squares gives $\mathbf{Y} - \hat{\mathbf{Y}} \perp \hat{\mathbf{Y}}$,

$$\|\mathbf{Y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2.$$

Thus

$$\text{RSS} = \|\mathbf{Y}\|^2 - \|\hat{\mathbf{Y}}\|^2.$$

Substituting,

$$\text{RSS} = \sum_i Y_i^2 - n\bar{Y}^2 - \sum_{j=1}^p \frac{((\mathbf{x}^{(j)})^\top \mathbf{Y})^2}{\|\mathbf{x}^{(j)}\|^2}.$$

Since

$$\sum_i (Y_i - \bar{Y})^2 = \sum_i Y_i^2 - n\bar{Y}^2,$$

we obtain

$$\text{RSS} = \sum_{i=1}^n (Y_i - \bar{Y})^2 - \sum_{j=1}^p \frac{((\mathbf{x}^{(j)})^\top \mathbf{Y})^2}{\|\mathbf{x}^{(j)}\|^2}.$$

Key idea. Orthogonality makes the regression decomposition literally a sum of independent squared projection lengths: each regressor explains a separate component of the centered response.

3 A3. Why orthogonal regressors minimize the total variance

Question (as assigned)

3. Consider the regression model in (1), and let the design matrix satisfy the following orthogonality property:

$$\mathbf{1}^\top \mathbf{x}^{(j)} = 0, \quad \|\mathbf{x}^{(j)}\| = c, \quad j = 1, \dots, p.$$

Show that

$$\sum_{j=1}^p \text{Var}(\hat{\beta}_{j,OLS})$$

is minimized when the covariates are orthogonal (as in Q.2).

Required theory from the lecture notes

For full column rank,

$$\text{Var}(\hat{\boldsymbol{\beta}}_{OLS}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

Thus

$$\text{Var}(\hat{\beta}_{j,OLS}) = \sigma^2 \left[(\mathbf{X}^\top \mathbf{X})^{-1} \right]_{j+1,j+1},$$

where the shift by one is because the first column corresponds to the intercept.

Solution

Because every covariate is centered, the intercept is orthogonal to the slope columns. Hence

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} n & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{G} \end{bmatrix}, \quad G_{jj} = c^2, \quad G_{jk} = \mathbf{x}^{(j)\top} \mathbf{x}^{(k)}.$$

Therefore

$$\text{Var}(\hat{\boldsymbol{\beta}}_{1:p}) = \sigma^2 \mathbf{G}^{-1}.$$

Consequently,

$$\sum_{j=1}^p \text{Var}(\hat{\beta}_j) = \sigma^2 \text{tr}(\mathbf{G}^{-1}).$$

The diagonal elements of $\mathbf{G}$ are fixed at c^2 , and by Cauchy–Schwarz / spectral inequalities the trace of the inverse is minimized when the eigenvalues of $\mathbf{G}$ are as equal as possible. Since

$$\text{tr}(\mathbf{G}) = pc^2$$

is fixed, the most balanced possible spectrum is

$$\lambda_1 = \dots = \lambda_p = c^2.$$

That happens exactly when

$$\mathbf{G} = c^2 \mathbf{I}_p,$$

i.e. when

$$\mathbf{x}^{(j)\top} \mathbf{x}^{(k)} = 0 \quad (j \neq k).$$

In that case,

$$\mathbf{G}^{-1} = \frac{1}{c^2} \mathbf{I}_p,$$

so

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{c^2}$$

for every j , and the sum equals

$$\sum_{j=1}^p \text{Var}(\hat{\beta}_j) = \frac{p\sigma^2}{c^2}.$$

Thus

$$\sum_{j=1}^p \text{Var}(\hat{\beta}_{j,OLS}) \text{ is minimized when the covariates are pairwise orthogonal.}$$

Intuition. Correlation among regressors makes the columns of the design matrix point in similar directions. Then the inverse of $\mathbf{X}^\top \mathbf{X}$ becomes inflated. Orthogonal columns avoid this competition and give the smallest total uncertainty for fixed regressor lengths.

4 A4. Gauss–Markov theorem for an arbitrary linear combination

Question (as assigned)

4. Let the design matrix $\mathbf{X}$ have rank $p + 1$. Then show that, for any parameter of the form $\mathbf{c}^\top \boldsymbol{\beta}$, the estimator $\mathbf{c}^\top \hat{\boldsymbol{\beta}}_{OLS}$ is the best linear unbiased estimator.

Required theory from the lecture notes

The notes state the key BLUE result:

$$\mathbb{E}(\hat{\boldsymbol{\beta}}_{OLS}) = \boldsymbol{\beta}, \quad \text{Var}(\hat{\boldsymbol{\beta}}_{OLS}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1},$$

and for any linear parametric function $\mathbf{c}^\top \boldsymbol{\beta}$, the estimator

$$\mathbf{c}^\top \hat{\boldsymbol{\beta}}_{OLS}$$

is BLUE. The proof is based on comparing it with an arbitrary linear unbiased estimator and showing that the variance difference is nonnegative.

Solution

Let

$$\theta = \mathbf{c}^\top \boldsymbol{\beta}.$$

The OLS estimator is

$$\hat{\theta}_{OLS} = \mathbf{c}^\top \hat{\boldsymbol{\beta}}_{OLS} = \mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

This is linear in $\mathbf{Y}$.

Step 1: unbiasedness. Since $\mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$,

$$\begin{aligned} \mathbb{E}(\hat{\theta}_{OLS}) &= \mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}(\mathbf{Y}) \\ &= \mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} \\ &= \mathbf{c}^\top \boldsymbol{\beta} \\ &= \theta. \end{aligned}$$

Hence $\hat{\theta}_{OLS}$ is unbiased.

Step 2: take an arbitrary linear unbiased estimator. Let

$$\tilde{\theta} = \mathbf{a}^\top \mathbf{Y}$$

be any linear unbiased estimator of θ . Unbiasedness means

$$\mathbb{E}(\tilde{\theta}) = \mathbf{a}^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{c}^\top \boldsymbol{\beta}$$

for every $\boldsymbol{\beta}$. Therefore

$$\mathbf{a}^\top \mathbf{X} = \mathbf{c}^\top.$$

Step 3: decompose $\mathbf{a}$. Let

$$\mathbf{a}_0 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{c}$$

be the coefficient vector that reproduces $\mathbf{c}$ on the column space, and write

$$\mathbf{a} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c} + \mathbf{d},$$

where $\mathbf{X}^\top \mathbf{d} = \mathbf{0}$. This decomposition is possible because

$$\mathbf{X}^\top (\mathbf{a} - \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}) = \mathbf{c} - \mathbf{c} = \mathbf{0}.$$

Step 4: compare variances. Because $\text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}$,

$$\text{Var}(\tilde{\theta}) = \sigma^2 \|\mathbf{a}\|^2.$$

Also,

$$\text{Var}(\hat{\theta}_{OLS}) = \sigma^2 \mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}.$$

Using orthogonality of the two components,

$$\|\mathbf{a}\|^2 = \left\| \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c} \right\|^2 + \|\mathbf{d}\|^2.$$

Now

$$\begin{aligned} \left\| \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c} \right\|^2 &= \mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c} \\ &= \mathbf{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}. \end{aligned}$$

Thus

$$\text{Var}(\tilde{\theta}) = \text{Var}(\hat{\theta}_{OLS}) + \sigma^2 \|\mathbf{d}\|^2 \geq \text{Var}(\hat{\theta}_{OLS}).$$

Equality occurs only when $\mathbf{d} = \mathbf{0}$, which gives the OLS estimator.

Therefore,

$$\boxed{\mathbf{c}^\top \hat{\boldsymbol{\beta}}_{OLS} \text{ is the BLUE of } \mathbf{c}^\top \boldsymbol{\beta}.$$

5 A5. Variance of $\hat{\beta}_1$ with two centered regressors

Question (as assigned)

5. Let

$$Y_i = \beta_0 + \beta_1(x_{i1} - \bar{x}_1) + \beta_2(x_{i2} - \bar{x}_2) + \varepsilon_i, \quad i = 1, 2, \dots, n,$$

where

$$\bar{x}_j = \frac{\sum_{i=1}^n x_{ij}}{n}, \quad \mathbb{E}(\varepsilon) = \mathbf{0}, \quad \text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

If $\hat{\beta}_1$ is the least squares estimate of β_1 , show that

$$\text{Var}[\hat{\beta}_1] = \frac{\sigma^2}{\sum_i (x_{i1} - \bar{x}_1)^2 (1 - r^2)},$$

where r is the correlation coefficient of the n pairs (x_{i1}, x_{i2}) .

Required theory from the lecture notes

For a centered regression with two regressors,

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

After centering, the intercept is orthogonal to both regressors, so the slope covariance is obtained by inverting the 2×2 cross-product matrix.

Solution

Define the centered cross-products

$$S_{11} = \sum_i (x_{i1} - \bar{x}_1)^2, \quad S_{22} = \sum_i (x_{i2} - \bar{x}_2)^2, \\ S_{12} = \sum_i (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2).$$

The correlation is

$$r = \frac{S_{12}}{\sqrt{S_{11}S_{22}}}.$$

The slope block of $\mathbf{X}^\top \mathbf{X}$ is

$$\mathbf{G} = \begin{bmatrix} S_{11} & S_{12} \\ S_{12} & S_{22} \end{bmatrix}.$$

The inverse of a 2×2 matrix is

$$\mathbf{G}^{-1} = \frac{1}{S_{11}S_{22} - S_{12}^2} \begin{bmatrix} S_{22} & -S_{12} \\ -S_{12} & S_{11} \end{bmatrix}.$$

Therefore the (1,1) element is

$$(\mathbf{G}^{-1})_{11} = \frac{S_{22}}{S_{11}S_{22} - S_{12}^2}.$$

Hence

$$\text{Var}(\hat{\beta}_1) = \sigma^2 \frac{S_{22}}{S_{11}S_{22} - S_{12}^2}.$$

Factor $S_{11}S_{22}$ from the denominator:

$$S_{11}S_{22} - S_{12}^2 = S_{11}S_{22} \left(1 - \frac{S_{12}^2}{S_{11}S_{22}} \right).$$

But

$$\frac{S_{12}^2}{S_{11}S_{22}} = r^2.$$

Thus

$$\text{Var}(\hat{\beta}_1) = \sigma^2 \frac{S_{22}}{S_{11}S_{22}(1 - r^2)} = \frac{\sigma^2}{S_{11}(1 - r^2)}.$$

Finally,

$$\boxed{\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{[\sum_i (x_{i1} - \bar{x}_1)^2] (1 - r^2)}}.$$

Interpretation. There are two separate sources of precision:

- larger spread in x_1 (larger S_{11}) reduces the variance;
- stronger correlation between x_1 and x_2 makes $1 - r^2$ small and inflates the variance.

This is the algebraic expression of multicollinearity for the two-regressor case.

6 A6. The minimum-variance nonnegative quadratic unbiased estimator of error variance

Question (as assigned)

6. Let $Y_1, Y_2, \dots, Y_n$ be n independent random variables with common variance σ^2 and common third and fourth moments, μ_3 and μ_4 , respectively, about their means. If $E[Y] = X\beta$, where X is an $n \times p$ matrix of rank p , then $(n-p)S^2$ is the unique nonnegative quadratic unbiased estimate of $(n-p)\sigma^2$ with minimum variance when $\mu_4 = 3\sigma^4$ or when the diagonal elements of P are all equal. [Hint: See Seber and Lee, Chapter 3, Section 3.3]

Required theory from the lecture notes

The lecture notes define the projection matrix

$$P_X = X(X^\top X)^{-1}X^\top$$

and the residual maker

$$M_X = I - P_X.$$

Then M_X is symmetric and idempotent, with

$$\text{rank}(M_X) = n - p, \quad \text{tr}(M_X) = n - p.$$

The residual sum of squares is

$$\text{RSS} = Y^\top M_X Y.$$

The notes also prove

$$E\left(\frac{\text{RSS}}{n-p}\right) = \sigma^2.$$

Important notation clarification

The assignment says “diagonal elements of P are all equal.” In the lecture notes, P_X denotes the *hat* projection and $I - P_X$ denotes the residual projection. Different texts use P for different projection matrices. To avoid ambiguity, this solution writes the residual projection as M .

Solution: identify the candidate

The usual unbiased variance estimator is

$$S^2 = \frac{\text{RSS}}{n-p} = \frac{Y^\top M Y}{n-p}.$$

Thus

$$(n-p)S^2 = Y^\top M Y.$$

Since $Y = X\beta + \varepsilon$ and $MX = 0$,

$$Y^\top M Y = \varepsilon^\top M \varepsilon.$$

So the candidate is a quadratic form in the errors.

Step 1: unbiasedness

For any random vector ε with mean zero and covariance $\sigma^2 I$,

$$E(\varepsilon^\top M \varepsilon) = \sigma^2 \operatorname{tr}(M),$$

because $MX = 0$ removes the mean contribution. Since $\operatorname{tr}(M) = n - p$,

$$E\{(n - p)S^2\} = (n - p)\sigma^2.$$

Hence S^2 is unbiased for σ^2 .

Step 2: consider an arbitrary nonnegative quadratic unbiased estimator

Let

$$Q = Y^\top A Y,$$

where A is symmetric positive semidefinite. For Q to estimate only the error variance, the part of Y lying in $\mathcal{C}(X)$ must not contribute, so require

$$AX = 0.$$

Unbiasedness for $(n - p)\sigma^2$ requires

$$\operatorname{tr}(A) = n - p.$$

For independent observations with common fourth central moment μ_4 , the variance of a quadratic form is

$$\operatorname{Var}(\varepsilon^\top A \varepsilon) = 2\sigma^4 \operatorname{tr}(A^2) + (\mu_4 - 3\sigma^4) \sum_{i=1}^n a_{ii}^2.$$

This is the key fourth-moment expression behind the result quoted in the question.

Step 3: the normal-kurtosis case $\mu_4 = 3\sigma^4$

When

$$\mu_4 = 3\sigma^4,$$

the second term disappears, giving

$$\operatorname{Var}(Q) = 2\sigma^4 \operatorname{tr}(A^2).$$

Therefore we must minimize $\operatorname{tr}(A^2)$ subject to

$$A \succeq 0, \quad AX = 0, \quad \operatorname{tr}(A) = n - p.$$

The condition $AX = 0$ means that the range of A is contained in the $(n - p)$ -dimensional residual space. Let the nonzero eigenvalues of A be $\lambda_1, \dots, \lambda_r$, with $r \leq n - p$. Then

$$\operatorname{tr}(A) = \sum_{j=1}^r \lambda_j = n - p,$$

and

$$\operatorname{tr}(A^2) = \sum_{j=1}^r \lambda_j^2.$$

For fixed sum, the sum of squares is minimized when all nonzero eigenvalues are equal. Since we also have $r \leq n - p$, the smallest value occurs at

$$r = n - p, \quad \lambda_1 = \cdots = \lambda_{n-p} = 1.$$

Hence

$$A = M,$$

the orthogonal projection onto the residual space.

Therefore M is the unique minimizer, and

$$Q = Y^T M Y = \text{RSS}.$$

Thus

$$S^2 = \frac{\text{RSS}}{n - p}$$

What the equal-diagonal condition is saying

For general μ_4 , the second term in the variance matters. The standard quadratic-estimation result quoted in the assignment (with the stated equal-diagonal condition) gives the same uniqueness conclusion in that special design situation. The equality-of-diagonals condition removes an additional source of variation caused by unequal diagonal weights in the quadratic form. The detailed fourth-moment argument above is the essential mechanism; the assignment explicitly points to Seber and Lee, Chapter 3, Section 3.3 for the general theorem.

Part II

B: Problems with normality assumption on errors

Throughout Part B we additionally assume the normal linear regression model. The assignment literally prints $\varepsilon_i \stackrel{iid}{\sim} N(0, 1)$; the derivations below are written for the general σ^2 case, and $\sigma^2 = 1$ follows immediately by substitution.

7 B1. Independence of $\hat{\beta}_{OLS}$ and RSS

Question (as assigned)

1. Show that $\hat{\beta}_{OLS}$ and RSS are independent.

Required theory from the lecture notes

Under normality,

$$\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}).$$

The OLS estimator and residual are

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T Y, \quad \mathbf{e} = (I - P_X)Y.$$

The notes use the normal-model geometry together with the fact that orthogonal Gaussian components are independent.

Solution

Write

$$P = P_X = X(X^T X)^{-1} X^T, \quad M = I - P.$$

Then

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T Y$$

is a linear function of Y , while

$$\mathbf{e} = MY$$

is another linear function.

To establish independence, it is enough under joint normality to show that their covariance is zero.

Step 1: compute the covariance.

$$\begin{aligned} \text{Cov}(\hat{\beta}_{OLS}, \mathbf{e}) &= (X^T X)^{-1} X^T \text{Var}(Y) M \\ &= (X^T X)^{-1} X^T (\sigma^2 I) M \\ &= \sigma^2 (X^T X)^{-1} X^T M. \end{aligned}$$

But

$$X^T M = X^T (I - P) = X^T - X^T P.$$

Since

$$X^T P = X^T X (X^T X)^{-1} X^T = X^T,$$

we get

$$X^T M = 0.$$

Therefore

$$\text{Cov}(\hat{\beta}_{OLS}, \mathbf{e}) = 0.$$

Step 2: use joint normality. Both $\hat{\beta}_{OLS}$ and $\mathbf{e}$ are linear transformations of a multivariate normal vector Y . Hence they are jointly normal. Joint normality plus zero covariance implies independence.

Step 3: RSS is a function of the residual.

$$\text{RSS} = \mathbf{e}^T \mathbf{e}.$$

Since $\hat{\beta}_{OLS}$ is independent of the entire residual vector $\mathbf{e}$, it is also independent of every measurable function of $\mathbf{e}$, in particular RSS.

Therefore,

$$\boxed{\hat{\beta}_{OLS} \perp\!\!\!\perp \text{RSS} .}$$

Geometric intuition. OLS extracts the component of Y lying in $\mathcal{C}(X)$, while the residual extracts the orthogonal component in $\mathcal{C}(X)^\perp$. For a multivariate normal vector, orthogonal components are independent.

8 B2. Consistency of the MLEs $(\hat{\beta}_{ML}, \hat{\sigma}_{ML}^2)$

Question (as assigned)

2. Show that both $\hat{\beta}_{ML}$ and $\hat{\sigma}_{ML}^2$ are consistent estimators.

Required theory from the lecture notes

The normal-model log-likelihood is

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (Y - X\beta)^T (Y - X\beta).$$

Maximization with respect to β is equivalent to least squares, so

$$\hat{\beta}_{ML} = \hat{\beta}_{OLS}.$$

The MLE of the variance is

$$\hat{\sigma}_{ML}^2 = \frac{\text{RSS}}{n}.$$

The unbiased estimator instead uses $\text{RSS} / (n - p - 1)$, so the MLE is biased at finite n but can still be consistent.

Consistency assumptions

For a standard fixed-design consistency argument assume

$$\frac{1}{n} X^T X \longrightarrow Q, \quad Q \text{ positive definite,}$$

and errors are independent with mean zero and finite variance (normality gives more than enough moment assumptions here).

Solution for $\hat{\beta}_{ML}$

Since $\hat{\beta}_{ML} = \hat{\beta}_{OLS}$,

$$\hat{\beta} - \beta = (X^T X)^{-1} X^T \epsilon.$$

Under the model,

$$E(\hat{\beta} - \beta) = 0,$$

and

$$\text{Var}(\hat{\beta} - \beta) = \sigma^2 (X^T X)^{-1}.$$

Now

$$X^T X = n \left(\frac{1}{n} X^T X \right),$$

so

$$(X^T X)^{-1} = \frac{1}{n} \left(\frac{1}{n} X^T X \right)^{-1}.$$

Because $n^{-1} X^T X \rightarrow Q$ with $Q \succ 0$,

$$\left(\frac{1}{n} X^T X \right)^{-1} \rightarrow Q^{-1},$$

and consequently

$$\text{Var}(\hat{\beta} - \beta) = O(n^{-1}) \rightarrow 0.$$

For each component, Chebyshev's inequality gives, for every $\epsilon > 0$,

$$P(|\hat{\beta}_j - \beta_j| > \epsilon) \leq \frac{\text{Var}(\hat{\beta}_j)}{\epsilon^2} \rightarrow 0.$$

Hence

$$\boxed{\hat{\beta}_{ML} \xrightarrow{p} \beta.}$$

Solution for $\hat{\sigma}_{ML}^2$

Recall

$$\hat{\sigma}_{ML}^2 = \frac{\text{RSS}}{n}.$$

Since

$$\text{RSS} = \epsilon^T M \epsilon, \quad M = I - P_X,$$

we have

$$E(\text{RSS}) = \sigma^2 \text{tr}(M) = \sigma^2(n - p - 1).$$

Therefore

$$E(\hat{\sigma}_{ML}^2) = \sigma^2 \left(1 - \frac{p+1}{n} \right) \rightarrow \sigma^2.$$

Moreover, under normality,

$$\frac{\text{RSS}}{\sigma^2} \sim \chi_{n-p-1}^2.$$

Thus

$$\frac{\text{RSS}}{n} = \frac{\sigma^2}{n} \chi_{n-p-1}^2.$$

Since a chi-square variable with ν degrees of freedom satisfies $\chi_{\nu}^2/\nu \rightarrow 1$ in probability as $\nu \rightarrow \infty$,

$$\frac{\text{RSS}}{n} = \sigma^2 \left(\frac{n-p-1}{n} \right) \left(\frac{\chi_{n-p-1}^2}{n-p-1} \right) \xrightarrow{p} \sigma^2.$$

Therefore

$$\hat{\sigma}_{ML}^2 \xrightarrow{p} \sigma^2.$$

Final conclusion:

$$\hat{\beta}_{ML} \xrightarrow{p} \beta, \quad \hat{\sigma}_{ML}^2 \xrightarrow{p} \sigma^2.$$

9 B3. LRTs for linear restrictions

Question (as assigned)

3. Under the normal linear regression model described above, consider the following problems of testing:

(a) Suppose there is a hierarchy among the regressors, so that we want to test

$$H_0 : \beta_1 - \beta_2 = \cdots = \beta_{p-1} - \beta_p = 1.$$

Find the LRT test statistic for this testing explicitly and its exact distribution.

(b) Suppose we want to test

$$H_0 : \beta_1 + \beta_2 + \beta_p = k.$$

Find the LRT test statistic for this testing explicitly and its exact distribution.

Required theory from the lecture notes

The LRT for

$$H_0 : A\beta = c \quad \text{versus} \quad H_1 : \text{no restriction}$$

is

$$\Lambda(Y) = \frac{L^*(\hat{\beta}_{H_0}, \hat{\sigma}_{H_0}^2)}{L^*(\hat{\beta}, \hat{\sigma}^2)}.$$

Under the normal model,

$$\hat{\sigma}^2 = \frac{\text{RSS}}{n}$$

and under the restriction

$$\hat{\sigma}_{H_0}^2 = \frac{\text{RSS}_{H_0}}{n}.$$

At the maximized likelihoods, the constants cancel and

$$\Lambda = \left(\frac{\text{RSS}}{\text{RSS}_{H_0}} \right)^{n/2}.$$

For q independent restrictions, the usual F statistic is

$$F = \frac{(\text{RSS}_{H_0} - \text{RSS})/q}{\text{RSS}/\nu}, \quad \nu = n - p - 1,$$

and under H_0 ,

$$F \sim F_{q,\nu}.$$

The lecture notes also write this as

$$\Lambda = \left(1 + \frac{q}{\nu} F \right)^{-n/2}.$$

B3(a): hierarchy restrictions

There are

$$q = p - 1$$

restrictions:

$$\beta_1 - \beta_2 = 1, \quad \beta_2 - \beta_3 = 1, \quad \dots, \quad \beta_{p-1} - \beta_p = 1.$$

These are linearly independent. Hence the restricted parameter space has $p - (p - 1) = 1$ free slope parameter (in addition to the intercept).

Represent the restrictions as

$$A\beta = c,$$

where A is a $(p - 1) \times (p + 1)$ matrix acting on

$$\beta = (\beta_0, \beta_1, \dots, \beta_p)^T,$$

with rows

$$(0, 1, -1, 0, \dots, 0), \quad (0, 0, 1, -1, \dots, 0), \quad \dots, \quad (0, 0, \dots, 0, 1, -1),$$

and $c = (1, 1, \dots, 1)^T$.

Let RSS_{H_0} denote the residual sum of squares from the restricted fit. Then

$$\Lambda = \left(\frac{\text{RSS}}{\text{RSS}_{H_0}} \right)^{n/2}.$$

Equivalently,

$$F = \frac{(\text{RSS}_{H_0} - \text{RSS})/(p - 1)}{\text{RSS}/(n - p - 1)} \sim F_{p-1, n-p-1} \quad (H_0).$$

Solving for Λ in terms of F ,

$$\text{RSS}_{H_0} = \text{RSS} \left(1 + \frac{p - 1}{n - p - 1} F \right),$$

so

$$\Lambda = \left(1 + \frac{p - 1}{n - p - 1} F_{p-1, n-p-1} \right)^{-n/2}.$$

The LRT rejects for small Λ , equivalently for large F .

B3(b): one linear restriction

The restriction

$$\beta_1 + \beta_2 + \beta_p = k$$

is a single restriction, hence $q = 1$. Take

$$A = (0, 1, 1, 0, \dots, 0, 1), \quad c = k.$$

Let RSS_{H_0} be the RSS of the restricted fit. Then

$$\Lambda = \left(\frac{\text{RSS}}{\text{RSS}_{H_0}} \right)^{n/2}.$$

The corresponding F statistic is

$$F = \frac{\text{RSS}_{H_0} - \text{RSS}}{\text{RSS}/(n - p - 1)} \sim F_{1, n-p-1} \quad (H_0).$$

Hence

$$\Lambda = \left(1 + \frac{1}{n-p-1} F_{1,n-p-1} \right)^{-n/2}.$$

Connection with a t test. Because

$$F_{1,\nu} = t_{\nu}^2,$$

a single linear restriction can equivalently be tested by a t statistic.

10 B4. LRT statistic and the simple correlation coefficient

Question (as assigned)

4. Consider the simple linear regression model

$$Y_i = \alpha + \beta x + \varepsilon_i, \quad i = 1, \dots, n,$$

and the problem of testing $H_0 : \beta = 0$. Let $\Lambda(Y)$ is the LRT test statistic for testing H_0 against H_1 : no restriction. Show that

$$r^2 = 1 - \{\Lambda(Y)\}^{2/n}.$$

The assignment further notes that this, together with the corresponding lecture-note results, gives the same relationship for the simple, multiple, and partial correlation coefficients.

Required theory from the lecture notes

Under $H_0 : \beta = 0$, the model is intercept-only. Therefore

$$\text{RSS}_{H_0} = \sum_{i=1}^n (Y_i - \bar{Y})^2 = \text{TSS}.$$

For the unrestricted simple regression,

$$R^2 = r^2 = 1 - \frac{\text{RSS}}{\text{TSS}}.$$

The normal-model LRT is

$$\Lambda = \left(\frac{\text{RSS}}{\text{RSS}_{H_0}} \right)^{n/2}.$$

Solution

Because $H_0 : \beta = 0$, the restricted model contains only the intercept:

$$Y_i = \alpha + \varepsilon_i.$$

Its least-squares fitted value is $\hat{\alpha} = \bar{Y}$, hence

$$\text{RSS}_{H_0} = \sum_i (Y_i - \bar{Y})^2 = \text{TSS}.$$

The unrestricted simple-regression RSS is RSS.

Therefore

$$\Lambda = \left(\frac{\text{RSS}}{\text{TSS}} \right)^{n/2}.$$

Raise both sides to the power $2/n$:

$$\Lambda^{2/n} = \frac{\text{RSS}}{\text{TSS}}.$$

But by definition of the simple-regression coefficient of determination,

$$r^2 = 1 - \frac{\text{RSS}}{\text{TSS}}.$$

Substituting the likelihood ratio expression gives

$$\boxed{r^2 = 1 - \Lambda^{2/n}}.$$

Interpretation. A good unrestricted fit makes RSS much smaller than TSS, hence Λ becomes small. At the same time r^2 becomes large. Thus the correlation coefficient and the LRT statistic carry the same information, but in opposite directions.

11 B5. Scheffé simultaneous confidence intervals for four slope combinations

Question (as assigned)

5. In the normal linear regression model as described above, consider the problem of finding the simultaneous confidence intervals (SCIs) for β_1 , β_2 , $\beta_1 + \beta_2$ and $\beta_1 - \beta_2$. Find Scheffé's SCI for the confidence coefficient $1 - \alpha$.

Required theory from the lecture notes

For r linearly independent linear combinations collected as

$$A\beta,$$

the lecture notes give the confidence ellipsoid

$$\frac{\nu}{r \text{ RSS}} (A\hat{\beta} - A\beta)^T [A(X^T X)^{-1} A^T]^{-1} (A\hat{\beta} - A\beta) \leq F_{1-\alpha; r, \nu},$$

where

$$\nu = n - p - 1.$$

Scheffé's method converts this ellipsoidal statement into simultaneous intervals. If

$$L = A(X^T X)^{-1} A^T,$$

then for the j th specified linear combination,²

$$a_j^T \beta \in a_j^T \hat{\beta} \pm \sqrt{\frac{r \text{ RSS}}{\nu} F_{1-\alpha; r, \nu} a_j^T (X^T X)^{-1} a_j}$$

simultaneously with confidence at least $1 - \alpha$.

²The supplied notes derive this via the supremum identity for a positive-definite quadratic form.

Solution

Only the two-dimensional slope parameter

$$\theta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$$

is involved. The four requested linear combinations are generated by

$$\mathbf{c}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{c}_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{c}_4 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

These four vectors lie in a two-dimensional space. Hence Scheffé must protect the entire two-dimensional family, so use

$$r = 2, \quad \nu = n - p - 1.$$

Let

$$V = [(X^T X)^{-1}]_{\{\beta_1, \beta_2\} \times \{\beta_1, \beta_2\}}$$

be the 2×2 slope covariance matrix without the factor σ^2 . Define

$$C_\alpha = \sqrt{\frac{2 \text{ RSS}}{\nu} F_{1-\alpha; 2, \nu}}.$$

Then the four simultaneous intervals are:

$$\boxed{\beta_1 \in \hat{\beta}_1 \pm C_\alpha \sqrt{V_{11}}}$$

$$\boxed{\beta_2 \in \hat{\beta}_2 \pm C_\alpha \sqrt{V_{22}}}$$

$$\boxed{\beta_1 + \beta_2 \in (\hat{\beta}_1 + \hat{\beta}_2) \pm C_\alpha \sqrt{V_{11} + V_{22} + 2V_{12}}}$$

$$\boxed{\beta_1 - \beta_2 \in (\hat{\beta}_1 - \hat{\beta}_2) \pm C_\alpha \sqrt{V_{11} + V_{22} - 2V_{12}}}.$$

These four statements hold simultaneously with confidence coefficient $1 - \alpha$.

Important point. The factor is based on $r = 2$, not $r = 4$, because the four contrasts are functions of only two underlying slope parameters. This is why the dimension of the parameter space being protected matters.

12 B6. General Scheffé interval and comparison with an individual interval

Question (as assigned)

6. Consider a linear combination of β , say $\mathbf{a}^\top \beta$. Find the Scheffé's confidence interval of the form

$$[\mathbf{a}^\top \hat{\beta}_{OLS} \pm G_\alpha(\mathbf{a})]$$

which is valid for all $\mathbf{a} \neq 0$. In particular, consider a given value of $\mathbf{a} = \mathbf{a}_0$. Find the individual confidence interval for $\mathbf{a}_0^\top \beta$ of the form

$$[\mathbf{a}_0^\top \hat{\beta}_{OLS} \pm H_\alpha(\mathbf{a}_0)].$$

Compare $G_\alpha(\mathbf{a}_0)$ and $H_\alpha(\mathbf{a}_0)$. Which confidence interval is wider and why?

Required theory from the lecture notes

For a fixed a , the exact individual confidence interval is

$$a^T \beta \in a^T \hat{\beta} \pm t_{1-\alpha/2, \nu} \sqrt{\frac{\text{RSS}}{\nu}} a^T (X^T X)^{-1} a, \quad \nu = n - p - 1.$$

The Scheffé result protects *all* linear combinations in the relevant r -dimensional space. If the entire parameter space of $\beta \in \mathbb{R}^{p+1}$ is protected, take $r = p + 1$.

Solution: Scheffé interval valid for all $a \neq 0$

Let

$$m = p + 1, \quad \nu = n - p - 1.$$

The Scheffé simultaneous inequality gives

$$a^T \beta \in a^T \hat{\beta}_{OLS} \pm \sqrt{\frac{m \text{ RSS}}{\nu} F_{1-\alpha; m, \nu}} a^T (X^T X)^{-1} a$$

for every $a \neq 0$ simultaneously. Therefore

$$G_\alpha(a) = \sqrt{\frac{(p+1) \text{ RSS}}{n-p-1} F_{1-\alpha; p+1, n-p-1}} a^T (X^T X)^{-1} a.$$

Solution: individual interval for a fixed a_0

For a single fixed linear combination, the standardized statistic is t with $\nu = n - p - 1$ degrees of freedom. Hence

$$a_0^T \beta \in a_0^T \hat{\beta}_{OLS} \pm t_{1-\alpha/2, \nu} \sqrt{\frac{\text{RSS}}{\nu}} a_0^T (X^T X)^{-1} a_0.$$

Thus

$$H_\alpha(a_0) = t_{1-\alpha/2, \nu} \sqrt{\frac{\text{RSS}}{\nu}} a_0^T (X^T X)^{-1} a_0.$$

Comparison

Both half-widths contain the same standard-error term

$$\sqrt{\frac{\text{RSS}}{\nu}} a_0^T (X^T X)^{-1} a_0.$$

The only difference is the multiplier:

$$\text{Scheffé: } \sqrt{m F_{1-\alpha; m, \nu}}, \quad \text{individual: } t_{1-\alpha/2, \nu}.$$

The Scheffé multiplier is larger because it must guarantee coverage over an entire continuum of directions a , whereas the individual interval protects one predetermined direction only.

Hence

$$G_\alpha(a_0) \geq H_\alpha(a_0), \quad \text{so Scheffé's interval is wider.}$$

Why? Simultaneous coverage costs statistical width. One fixed interval only has one chance to fail; Scheffé must guard against the worst direction in the entire parameter space.

13 B7. Confidence intervals along the largest and smallest eigenvalue directions

Question (as assigned)

7. Consider the problem of finding confidence intervals for two new observations $\mathbf{x}_0$ (known) and $\mathbf{z}_0$ (known), where $\mathbf{x}_0$ and $\mathbf{z}_0$ are unit eigenvectors corresponding to the highest and smallest eigenvalues of $X^\top X$, say $\ell_{\max}$ and $\ell_{\min}$ respectively, and $\ell_{\max}/\ell_{\min} = 10^4$. Interpret the result. Compare the widths of confidence intervals of $\mathbf{x}_0^\top \beta$ and $\mathbf{z}_0^\top \beta$.

Important note about the assignment text

The uploaded PS2 appears to contain an OCR/formula typo in the ratio: it prints the same symbol in numerator and denominator. The mathematically meaningful statement, consistent with the preceding words “highest and smallest eigenvalues”, is

$$\ell_{\max}/\ell_{\min} = 10^4.$$

The solution below uses this intended comparison.

Required theory from the lecture notes

For any fixed a ,

$$\text{Var}(a^\top \hat{\beta}) = \sigma^2 a^\top (X^\top X)^{-1} a.$$

If a is a unit eigenvector of $X^\top X$ with eigenvalue ℓ , then

$$(X^\top X)^{-1} a = \frac{1}{\ell} a,$$

and because $\|a\| = 1$,

$$a^\top (X^\top X)^{-1} a = \frac{1}{\ell}.$$

Therefore the standard error is proportional to $1/\sqrt{\ell}$.

Solution

Let

$$X^\top X \mathbf{x}_0 = \ell_{\max} \mathbf{x}_0, \quad X^\top X \mathbf{z}_0 = \ell_{\min} \mathbf{z}_0,$$

with

$$\|\mathbf{x}_0\| = \|\mathbf{z}_0\| = 1.$$

Then

$$\mathbf{x}_0^\top (X^\top X)^{-1} \mathbf{x}_0 = \frac{1}{\ell_{\max}},$$

while

$$\mathbf{z}_0^\top (X^\top X)^{-1} \mathbf{z}_0 = \frac{1}{\ell_{\min}}.$$

Hence the standard-error ratio is

$$\frac{\text{SE}(\mathbf{z}_0^\top \hat{\beta})}{\text{SE}(\mathbf{x}_0^\top \hat{\beta})} = \sqrt{\frac{1/\ell_{\min}}{1/\ell_{\max}}} = \sqrt{\frac{\ell_{\max}}{\ell_{\min}}}.$$

Using the assigned ratio,

$$\sqrt{10^4} = 100.$$

Therefore,

$$\boxed{\text{SE}(\mathbf{z}_0^T \hat{\beta}) = 100 \text{ SE}(\mathbf{x}_0^T \hat{\beta}).}$$

For confidence intervals using the same critical value, the widths are in the same ratio:

$$\boxed{\frac{\text{Width}(\mathbf{z}_0^T \beta)}{\text{Width}(\mathbf{x}_0^T \beta)} = 100.}$$

Interpretation. The design is far more informative in the direction $\mathbf{x}_0$ than in the direction $\mathbf{z}_0$. The largest-eigenvalue direction is a direction in parameter space that the design measures very strongly, while the smallest-eigenvalue direction is poorly identified. The latter can therefore have an interval 100 times wider.

Terminology note. The assignment calls these “confidence intervals for two new observations.” Because the requested quantities are explicitly $\mathbf{x}_0^T \beta$ and $\mathbf{z}_0^T \beta$, the calculation above is a confidence interval for the *mean linear function*. A predictive interval for a future response would additionally contain the new observation-error variance σ^2 .

Part III

C: Practical problems

Dataset instructions and reproducible analysis

The uploaded assignment specifies two datasets but does not embed the full data tables. Therefore, the practical section below gives the complete statistical workflow and reproducible code rather than inventing numerical coefficients. The Iris numerical values included later were obtained from the canonical 150-observation Iris table; the Boston section is written to execute directly once the assigned Boston Housing file is available as a CSV.

The code below is intentionally written so that the user can run it and regenerate every table, p-value, and figure.

Python packages

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf
from scipy import stats
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
```

14 C1. Separate simple regressions and significance tests

Question (as assigned)

1. Consider the Boston Housing (BH) dataset and the Iris dataset. For each dataset, carry out the following analyses: Consider separate simple linear regression models, with the response variable regressed on each explanatory variable individually. For each model, test whether the corresponding regressor has a statistically significant effect on the response.

Required theory

For a simple regression

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

testing whether the regressor has a statistically significant effect means

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_1 : \beta_1 \neq 0.$$

Under normality,

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} \sim t_{n-2} \quad (H_0).$$

The two-sided p-value is

$$p = 2P(T_{n-2} \geq |t|).$$

Reject H_0 at level α when $p < \alpha$.

Reproducible Python code

```
# Boston Housing: assume a CSV named BostonHousing.csv is present.
bh = pd.read_csv("BostonHousing.csv")

bh_response = "MEDV"
bh_predictors = [
    "CRIM", "ZN", "INDUS", "NOX", "RM", "AGE",
    "DIS", "RAD", "TAX", "PTRATIO", "B", "LSTAT"
]

simple_results_bh = []
for x in bh_predictors:
    fit = smf.ols(f"{bh_response} ~ {x}", data=bh).fit()
    simple_results_bh.append({
        "regressor": x,
        "coef": fit.params[x],
        "t": fit.tvalues[x],
        "p_value": fit.pvalues[x],
        "R2": fit.rsquared
    })

simple_results_bh = pd.DataFrame(simple_results_bh)
print(simple_results_bh)

# Iris
iris_raw = load_iris(as_frame=True)
iris = iris_raw.frame.rename(columns={
    "sepal length (cm)": "Sepal_Length",
    "sepal width (cm)": "Sepal_Width",
    "petal length (cm)": "Petal_Length",
    "petal width (cm)": "Petal_Width"
})

iris_predictors = ["Sepal_Width", "Petal_Length", "Petal_Width"]

simple_results_iris = []
for x in iris_predictors:
    fit = smf.ols(f"Sepal_Length ~ {x}", data=iris).fit()
    simple_results_iris.append({
        "regressor": x,
        "coef": fit.params[x],
        "t": fit.tvalues[x],
        "p_value": fit.pvalues[x],
        "R2": fit.rsquared
    })

simple_results_iris = pd.DataFrame(simple_results_iris)
print(simple_results_iris)
```

Canonical Iris numerical results

For the canonical Iris data, the separate simple regressions give approximately:

Regressor	Slope	t	p -value	R^2
Sepal Width	-0.22336	-1.4403	0.1519	0.01382
Petal Length	0.40892	21.6460	1.04×10^{-47}	0.75996
Petal Width	0.88858	17.2965	2.33×10^{-37}	0.66903

At the conventional 5% level, Petal Length and Petal Width are highly significant in the separate simple regressions, whereas Sepal Width is not.

15 C2. Multiple regression, individual effects, joint effect, R^2 , and equal-effect test

Question (as assigned)

2. Consider a multiple linear regression model in which the response variable is regressed simultaneously on all the explanatory variables. Perform the following two types of hypothesis tests:

- For each regressor individually, test whether it has a statistically significant effect on the response after accounting for the effects of all the other regressors. In other words, for a fixed $j = 1, \dots, p$, test $H_0 : \beta_j = 0$ against the appropriate alternative, with no restrictions imposed on β_k for $k \neq j$.
- Test whether the regressors, taken together, have a statistically significant effect on the response. That is, test $H_0 : \boldsymbol{\beta} = \mathbf{0}$, where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$. Find the coefficient of determination, and conclude on the strength of association of the response and the covariates.
- Consider the problem of testing that all regressors (except the intercept) have same effect on the response. What is the estimate of the common regression coefficient β and the associated p-value?

Required theory

For a single coefficient,

$$t_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}, \quad df = n - p - 1,$$

and under $H_0 : \beta_j = 0$,

$$t_j \sim t_{n-p-1}.$$

For the joint null

$$H_0 : \beta_1 = \dots = \beta_p = 0,$$

the restricted model has only the intercept. The F statistic is

$$F = \frac{(\text{RSS}_{H_0} - \text{RSS})/p}{\text{RSS}/(n - p - 1)} \sim F_{p, n-p-1}.$$

The multiple coefficient of determination is

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}.$$

For the equal-effect test, $H_0 : \beta_1 = \dots = \beta_p = b$, fit the restricted model using the single predictor

$$Z_i = x_{i1} + \dots + x_{ip}.$$

Then

$$Y_i = \beta_0 + bZ_i + \varepsilon_i$$

implements the constraint that all original slopes are equal to the same b . This gives a direct restricted RSS, after which the standard nested-model F test applies with $q = p - 1$ restrictions.

Reproducible code for all three tests

```
def equal_effect_test(df, y_col, predictors, standardized=False):
    work = df.copy()
    cols = predictors.copy()

    if standardized:
        for x in cols:
            work[x] = (work[x] - work[x].mean()) / work[x].std(ddof=1)

    # Full model
    X_full = sm.add_constant(work[cols])
    full = sm.OLS(work[y_col], X_full).fit()

    # Equality restriction: all slopes are the same b
    work["Z_equal"] = work[cols].sum(axis=1)
    X_equal = sm.add_constant(work[["Z_equal"]])
    equal = sm.OLS(work[y_col], X_equal).fit()

    rss_full = np.sum(full.resid**2)
    rss_equal = np.sum(equal.resid**2)
    p = len(cols)
    df2 = int(full.df_resid)

    F = ((rss_equal - rss_full) / (p - 1)) / (rss_full / df2)
    p_value = stats.f.sf(F, p - 1, df2)

    return full, equal, F, p_value

# Example: Iris
full_iris = smf.ols(
    "Sepal_Length ~ Sepal_Width + Petal_Length + Petal_Width",
    data=iris
).fit()
print(full_iris.summary())

# Joint test H0: all slopes = 0
restricted_iris = smf.ols("Sepal_Length ~ 1", data=iris).fit()
F_joint, p_joint, df_num, df_den = full_iris.compare_f_test(restricted_iris)
print("Joint F =", F_joint, "p =", p_joint)
print("R^2 =", full_iris.rsquared)

# Equal-slope test
full_iris, equal_iris, F_equal, p_equal = equal_effect_test(
    iris,
    "Sepal_Length",
    iris_predictors,
    standardized=False
)
print("Equal-effect slope estimate =", equal_iris.params["Z_equal"])
print("Equal-effect F =", F_equal)
print("Equal-effect p =", p_equal)
```

Canonical Iris numerical results

The full multiple regression is approximately

$$\widehat{\text{Sepal Length}} = 1.855997 + 0.650837(\text{Sepal Width}) + 0.709132(\text{Petal Length}) - 0.556483(\text{Petal Width}).$$

The coefficientwise p-values are approximately

Regressor	p-value
Sepal Width	1.20×10^{-17}
Petal Length	7.66×10^{-25}
Petal Width	2.41×10^{-5}

so all three are individually significant in the multiple regression at the 5% level.

The overall fit has

$$R^2 \approx 0.858612,$$

and the overall F test is approximately

$$F \approx 295.54, \quad p \approx 8.59 \times 10^{-62}.$$

Thus the regressors are highly significant jointly and explain a very large fraction of the observed variation in Sepal Length.

For the equal-effect restriction in the original measurement units,

$$\hat{b} \approx 0.313520, \quad F \approx 30.619, \quad p \approx 7.86 \times 10^{-12}.$$

So the data strongly reject the hypothesis that all three slopes are the same.

16 C3. Comparing simple and multiple regression coefficients and p-values

Question (as assigned)

3. For each regressor, compare the regression coefficient and corresponding p-value obtained from:

- (a) the simple linear regression model,
- (b) the multiple linear regression model with the individual test of $H_0 : \beta_j = 0$.

Based on the testing results, answer:

- (a) Are the regression coefficients the same sign in both cases? If not, explain the possible reasons in view of the underlying dataset.
- (b) Are the resulting p-values of the same order of magnitude? If they differ substantially, explain the possible reasons for the differences by examining the relationships among the variables in the dataset.

Required theory

The lecture notes explicitly distinguish the two interpretations:

simple slope = marginal association,

while

multiple slope = conditional association holding the other covariates fixed.

The notes also discuss Simpson's paradox: if the other covariates are related both to the regressor and the response, the coefficient can change substantially and can even change sign.

Canonical Iris comparison

From C1 and C2:

Regressor	Simple		Multiple	
	slope	p	slope	p
Sepal Width	-0.22336	0.1519	+0.65084	1.20×10^{-17}
Petal Length	+0.40892	1.04×10^{-47}	+0.70913	7.66×10^{-25}
Petal Width	+0.88858	2.33×10^{-37}	-0.55648	2.41×10^{-5}

(a) Why can the sign change?

Yes: the canonical Iris results show both kinds of sign change.

Sepal Width. The marginal simple-regression coefficient is negative, but after controlling for Petal Length and Petal Width the coefficient becomes positive:

$$-0.223 \longrightarrow +0.651.$$

Petal Width. The marginal relationship is strongly positive,

$$+0.889,$$

but after adjusting for the other regressors it becomes negative,

$$-0.556.$$

The reason is not an algebraic error. The three measurements are strongly related to one another because they jointly describe flower size and shape. In a simple regression, a regressor can partially act as a proxy for information carried by the other correlated regressors. In a multiple regression, the coefficient asks a different question: what happens when that regressor changes while the other measured characteristics are held fixed? This is exactly the distinction emphasized in the lecture notes and is a standard route to Simpson-type sign reversal.

(b) Why can the p-values differ drastically?

The standard error of a multiple coefficient is affected by the geometry of all predictor columns. Correlation among predictors makes $(X^T X)^{-1}$ larger in relevant directions, which can inflate standard errors.

At the same time, controlling for other variables can remove irrelevant variation from the regressor-response association. Therefore, either of the following may happen:

1. the coefficient gets much more statistically significant after adjustment;
2. the coefficient becomes much less significant because its apparent simple-regression association was largely shared with other regressors.

For Iris, Sepal Width changes from nonsignificant in the simple regression ($p \approx 0.152$) to extremely significant in the multiple regression. Petal Width remains significant, but its p-value becomes much larger after conditioning on Petal Length and Sepal Width. Both changes are explained by the correlations among the predictor variables.

Exam takeaway: a simple regression and a multiple regression coefficient are not estimating the same scientific quantity unless the other regressors are appropriately irrelevant or orthogonal.

17 C4. Standardization and the equal-effect hypothesis

Question (as assigned)

4. Standardize the data by centering and scaling each explanatory variable. Perform the testing in Q.2(c). Are the results you obtain now same as those prior to standardization? If there are any changes, explain why. In particular, discuss the role of standardization and the scale of measurement of the variables in the context of the two datasets.

Required theory from the lecture notes

If each regressor is divided by its standard deviation, the new design matrix is

$$X_s = XD,$$

where D is diagonal. Since multiplication by a nonsingular diagonal matrix does not change the column space,

$$\mathcal{C}(X_s) = \mathcal{C}(X), \quad P(X_s) = P(X).$$

Therefore fitted values, residuals, and RSS remain unchanged. But the numerical regression coefficients change according to the scale transformation.

The crucial point for Q2(c)

The hypothesis “all slopes are equal” is a statement about slopes in the *chosen units*. Suppose the original model is

$$Y = \beta_0 + bX_1 + bX_2 + \cdots + bX_p + \varepsilon.$$

Now standardize

$$Z_j = \frac{X_j - \bar{X}_j}{s_j}.$$

A one-unit increase in Z_j corresponds to an s_j -unit increase in X_j . Therefore the standardized slope becomes

$$\beta_j^{(s)} = s_j \beta_j.$$

Thus equal slopes before standardization do not in general remain equal after standardization unless the original scales happen to satisfy the needed relationship.

Reproducible code

```
# Standardized equal-effect test for Iris
full_s, equal_s, F_equal_s, p_equal_s = equal_effect_test(
    iris,
    "Sepal_Length",
    iris_predictors,
    standardized=True
)

print("Standardized equal-effect slope:", equal_s.params["Z_equal"])
print("Standardized equal-effect F:", F_equal_s)
print("Standardized equal-effect p:", p_equal_s)
```

Canonical Iris result

After standardizing the three predictors using their sample standard deviations, the multiple coefficients become approximately

$$(0.283678, 1.251829, -0.424172),$$

so they are still clearly unequal. The constrained common standardized slope is approximately

$$\hat{b}_{std} = 0.390165,$$

and the equal-effect test gives approximately

$$F \approx 60.854, \quad p < 10^{-12}.$$

Thus the conclusion remains the same: the equal-effect hypothesis is decisively rejected.

But the reason is important. The numerical common-slope estimate and the F statistic can change because the restriction itself has changed in meaning after rescaling. The fitted full model does not change as a geometric projection, but “all slopes equal” is not a scale-invariant restriction.

Contrast: If the question instead asked for the joint significance of all regressors, standardization would leave the full-model RSS and intercept-only RSS unchanged, so the overall F statistic and its p-value would remain unchanged. The special sensitivity here arises because Q2(c) imposes a coefficient-equality restriction.

18 C5. Geometry of the specified ellipse

Question (as assigned)

5. Consider drawing an ellipse from the equation

$$(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top L (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) = c,$$

where

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}, \quad \hat{\boldsymbol{\beta}} = (2, 3)^\top, \\ L = \left(1 - 0.5^{|i-j|}\right), \quad i, j \in \{1, 2\},$$

is symmetric positive definite, and $c = 5$.

- (a) Find the transformation $\beta_1 \rightarrow X$ and $\beta_2 \rightarrow Y$, and constants (a, b) such that the above equation of the ellipse can be written as

$$X^2/a^2 + Y^2/b^2 = c.$$

- (b) Show that with respect to the $(X/a, Y/b)$ -coordinate system, the above equation represents a circle with center $(0, 0)$ and radius $\sqrt{c}$.
- (c) Hence, show that with respect to the (X, Y) -coordinate system, the above equation represents a standard ellipse (where the two axes are the natural axes) with center $(0, 0)$ and (a, b) as the major and minor axes.
- (d) Hence, show that with respect to the $(\beta_1 - \hat{\beta}_1, \beta_2 - \hat{\beta}_2)$ coordinate system the above equation represents a non-standard ellipse with center $(0, 0)$.
- (e) Hence, show that with respect to the (β_1, β_2) coordinate system the above equation represents a non-standard ellipse with center $(\hat{\beta}_1, \hat{\beta}_2)$ with same angle and axes as in (d).

Note on part (e). The uploaded assignment says center $(\hat{\beta}_1, \hat{\beta}_1)$, but the preceding definition and the geometry imply the center should be $(\hat{\beta}_1, \hat{\beta}_2) = (2, 3)$. This solution uses the mathematically consistent center $(2, 3)$.

Required theory

A quadratic-form ellipse

$$\mathbf{u}^T L \mathbf{u} = c$$

with L positive definite can be diagonalized by an orthogonal eigenvector transformation. If

$$L = U \Lambda U^\top, \quad \Lambda = \text{diag}(\lambda_1, \lambda_2),$$

and $\mathbf{u} = U \mathbf{q}$, then

$$q_1^2 \lambda_1 + q_2^2 \lambda_2 = c.$$

A translation changes only the center, not the axis directions or lengths.

Solution

First,

$$L = \begin{bmatrix} 1 & 1 - 0.5 \\ 1 - 0.5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}.$$

Define centered coordinates

$$u = \beta_1 - 2, \quad v = \beta_2 - 3.$$

Then

$$\begin{bmatrix} u & v \end{bmatrix} \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = 5,$$

so

$$u^2 + uv + v^2 = 5.$$

(a) Rotation to principal coordinates

The eigenvectors of L are proportional to

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

with eigenvalues

$$\lambda_1 = 1.5, \quad \lambda_2 = 0.5.$$

Hence define

$$X = \frac{u + v}{\sqrt{2}} = \frac{\beta_1 + \beta_2 - 5}{\sqrt{2}}$$

and

$$Y = \frac{u - v}{\sqrt{2}} = \frac{\beta_1 - \beta_2 + 1}{\sqrt{2}}.$$

The inverse transformation is

$$u = \frac{X + Y}{\sqrt{2}}, \quad v = \frac{X - Y}{\sqrt{2}}.$$

Now

$$\begin{aligned} u^2 + uv + v^2 &= \frac{(X + Y)^2}{2} + \frac{(X + Y)(X - Y)}{2} + \frac{(X - Y)^2}{2} \\ &= \frac{3}{2}X^2 + \frac{1}{2}Y^2. \end{aligned}$$

Therefore

$$\frac{3}{2}X^2 + \frac{1}{2}Y^2 = 5.$$

The assignment asks for the form

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} = c.$$

Hence

$$\frac{1}{a^2} = \frac{3}{2}, \quad \frac{1}{b^2} = \frac{1}{2},$$

so

$$a = \sqrt{\frac{2}{3}}, \quad b = \sqrt{2}.$$

(b) Circle in the scaled coordinates

Let

$$U = \frac{X}{a}, \quad V = \frac{Y}{b}.$$

Then

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} = c$$

becomes

$$U^2 + V^2 = c.$$

Therefore the set is a circle centered at the origin with radius

$$\sqrt{c} = \sqrt{5}.$$

Thus

$$\text{In } (X/a, Y/b) \text{ coordinates: a circle of radius } \sqrt{5}.$$

(c) Ellipse in the (X, Y) coordinates

Undoing the unequal scaling gives

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} = c.$$

Dividing by c yields the usual standard-ellipse form

$$\frac{X^2}{a^2 c} + \frac{Y^2}{b^2 c} = 1.$$

Thus the actual semi-axis lengths are

$$a\sqrt{c} = \sqrt{10/3}, \quad b\sqrt{c} = \sqrt{10}.$$

Since $b > a$, the major axis is the Y direction and the minor axis is the X direction.

(d) Ellipse in the centered $(\beta_1 - \hat{\beta}_1, \beta_2 - \hat{\beta}_2)$ coordinates

In the centered coordinates

$$(u, v) = (\beta_1 - 2, \beta_2 - 3),$$

the equation is

$$u^2 + uv + v^2 = 5.$$

Because the cross term uv is present, the axes are not aligned with the original (u, v) coordinate axes. Therefore it is a non-standard ellipse, centered at

$$(0, 0).$$

Its natural principal axes are the eigenvector directions

$$(1, 1), \quad (1, -1).$$

(e) Ellipse in the original (β_1, β_2) coordinates

The only difference between (u, v) and (β_1, β_2) is translation by $(2, 3)$. Therefore the shape, angle, and axis lengths do not change. Only the center moves:

$$(\beta_1, \beta_2) = (2, 3).$$

Thus the ellipse in the original coordinates is the same non-standard ellipse translated so that its center is $(2, 3)$.

19 C6. Numerical confidence regions for (β_1, β_2)

Question (as assigned)

6. Consider a data set with $n = 20$ observations on (y, x_1, x_2) with

$$\begin{aligned} \sum_i x_{1,i} &= 1.8, & \sum_i x_{1,i}^2 &= 21.2, & \sum_i x_{2,i} &= 14.5, & \sum_i x_{2,i}^2 &= 83.7, \\ \sum_i y_i &= -66.6, & \sum_i x_{1,i}x_{2,i} &= -4, & \sum_i x_{1,i}y_i &= 34.6, & \sum_i x_{2,i}y_i &= -127.1. \end{aligned}$$

Further, if we fit a normal linear regression model

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \varepsilon_i, \quad i = 1, \dots, 20,$$

then the RSS would be 576.2. Suppose we are interested in finding simultaneous confidence interval for (β_1, β_2) . In the same plot, draw the following choices of simultaneous confidence intervals (SCI):

- (a) The SCI obtained by the smallest rectangle containing the confidence ellipsoid. (Note that in this case the confidence ellipsoid is approximately of the form $X^2/a^2 + Y^2/b^2 = c^2$ after centering. So smallest rectangle can be easily obtained.)
- (b) The SCI obtained by Bonferroni's method.
- (c) The SCI obtained by Scheffé's method.

Which of these three methods provide the best result?

Required theory

For slope vector $\theta = (\beta_1, \beta_2)^T$, let

$$G = Z_c^T Z_c$$

be the centered slope cross-product matrix. Then

$$\hat{\theta} = G^{-1}s,$$

where

$$s = \begin{bmatrix} \sum_i (x_{1,i} - \bar{x}_1)(y_i - \bar{y}) \\ \sum_i (x_{2,i} - \bar{x}_2)(y_i - \bar{y}) \end{bmatrix}.$$

The slope covariance matrix is

$$\text{Var}(\hat{\theta}) = \sigma^2 G^{-1}.$$

The 95% confidence ellipse for the two-dimensional slope parameter is obtained from the Scheffé/ellipsoidal inequality

$$(\hat{\theta} - \theta)^T G (\hat{\theta} - \theta) \leq \frac{2 \text{RSS}}{\nu} F_{0.95; 2, \nu}, \quad \nu = n - p - 1.$$

For a coordinatewise interval from the ellipse, the half-width for coordinate j is

$$\sqrt{c(G^{-1})_{jj}},$$

where

$$c = \frac{2 \text{RSS}}{\nu} F_{0.95;2,\nu}.$$

Step 1: compute the centered sufficient statistics

The sample means are

$$\bar{x}_1 = \frac{1.8}{20} = 0.09, \quad \bar{x}_2 = \frac{14.5}{20} = 0.725, \quad \bar{y} = \frac{-66.6}{20} = -3.33.$$

The centered cross-products are

$$\begin{aligned} S_{11} &= \sum x_1^2 - n\bar{x}_1^2 = 21.2 - 20(0.09)^2 = 21.038, \\ S_{22} &= \sum x_2^2 - n\bar{x}_2^2 = 83.7 - 20(0.725)^2 = 73.1875, \\ S_{12} &= \sum x_1 x_2 - n\bar{x}_1 \bar{x}_2 = -4 - 20(0.09)(0.725) = -5.305. \end{aligned}$$

Likewise,

$$\begin{aligned} S_{1y} &= \sum x_1 y - n\bar{x}_1 \bar{y} = 34.6 - 20(0.09)(-3.33) = 40.594, \\ S_{2y} &= \sum x_2 y - n\bar{x}_2 \bar{y} = -127.1 - 20(0.725)(-3.33) = -78.815. \end{aligned}$$

Therefore

$$G = \begin{bmatrix} 21.038 & -5.305 \\ -5.305 & 73.1875 \end{bmatrix}.$$

Step 2: invert the slope cross-product matrix

The determinant is

$$\det(G) = 21.038(73.1875) - (-5.305)^2.$$

Numerically,

$$G^{-1} \approx \begin{bmatrix} 0.04841802 & 0.00350958 \\ 0.00350958 & 0.01391793 \end{bmatrix}.$$

Step 3: calculate the slope estimates

$$\hat{\theta} = G^{-1} \begin{bmatrix} 40.594 \\ -78.815 \end{bmatrix} \approx \begin{bmatrix} 1.688873 \\ -0.954473 \end{bmatrix}.$$

Hence

$$\boxed{\hat{\beta}_1 \approx 1.68887, \quad \hat{\beta}_2 \approx -0.95447.}$$

Step 4: degrees of freedom and variance estimate

Here

$$n = 20, \quad p = 2, \quad \nu = n - p - 1 = 17.$$

Thus

$$\hat{\sigma}_{unb}^2 = \frac{576.2}{17} \approx 33.89412.$$

Step 5: confidence-ellipsoid constant

For a 95% confidence set and $r = 2$,

$$F_{0.95;2,17} \approx 3.59153.$$

Therefore

$$c = \frac{2 \cdot 576.2}{17}(3.59153) \approx 243.46352.$$

The ellipsoid is

$$(\theta - \hat{\theta})^T G(\theta - \hat{\theta}) \leq 243.46352.$$

(a) Smallest rectangle containing the confidence ellipse

For an ellipsoid

$$(\theta - \hat{\theta})^T G(\theta - \hat{\theta}) \leq c,$$

the extreme deviation in coordinate j is

$$\sqrt{c(G^{-1})_{jj}}.$$

Therefore

$$\Delta_1 = \sqrt{243.46352(0.04841802)} \approx 3.43337,$$

$$\Delta_2 = \sqrt{243.46352(0.01391793)} \approx 1.84079.$$

Hence the smallest containing rectangle is

$$\beta_1 \in [1.68887 - 3.43337, 1.68887 + 3.43337] = [-1.74450, 5.12224],$$

$$\beta_2 \in [-0.95447 - 1.84079, -0.95447 + 1.84079] = [-2.79526, 0.88632].$$

(b) Bonferroni simultaneous intervals

We have two quantities, so use individual confidence coefficient

$$1 - \frac{\alpha}{2} = 0.975$$

for each interval. Thus the critical value is

$$t_{1-\alpha/(2r),\nu} = t_{0.9875,17} \approx 2.45805.$$

The standard errors are

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{576.2}{17}(0.04841802)} \approx 1.28056,$$

$$\text{SE}(\hat{\beta}_2) = \sqrt{\frac{576.2}{17}(0.01391793)} \approx 0.68676.$$

Thus the Bonferroni half-widths are

$$2.45805(1.28056) \approx 3.14888,$$

$$2.45805(0.68676) \approx 1.68826.$$

Therefore

$$\beta_1 \in [-1.46001, 4.83776],$$

$$\beta_2 \in [-2.64274, 0.73379].$$

(c) Scheffé simultaneous intervals

For the two slope parameters, Scheffé uses $r = 2$. Thus the multiplier is

$$\sqrt{\frac{2 \text{RSS}}{\nu} F_{0.95;2,17}} = \sqrt{c}.$$

Therefore the Scheffé coordinate intervals are exactly the coordinate projections of the confidence ellipse. Consequently their half-widths are

$$3.43337, \quad 1.84079,$$

and the intervals are exactly the same as in part (a):

$$\beta_1 \in [-1.74450, 5.12224],$$

$$\beta_2 \in [-2.79526, 0.88632].$$

Why Scheffé equals the ellipse projection here

The confidence ellipsoid itself is the exact two-dimensional Scheffé region for the two-dimensional slope parameter space. Taking the coordinate projections of that ellipsoid produces exactly the corresponding Scheffé coordinate intervals. Thus no discrepancy is expected between the smallest axis-aligned rectangle containing the ellipse and the coordinatewise Scheffé intervals.

Numerical comparison

Method	Half-width for β_1	Half-width for β_2	Relative width
Ellipse projection / smallest rectangle	3.43337	1.84079	wider
Bonferroni	3.14888	1.68826	narrower
Scheffé	3.43337	1.84079	wider

Thus, for the requested coordinatewise SCI widths,

Bonferroni gives the narrowest intervals of the three.

However, there is an important qualification: the confidence *ellipse* contains more geometric information than an axis-aligned rectangle, because it retains the dependence between $\hat{\beta}_1$ and $\hat{\beta}_2$. So if “best” means most informative geometric confidence region, the ellipse itself is preferable; if it means shortest separate simultaneous coordinate intervals here, Bonferroni is preferable.

Python code to reproduce the numerical calculation and plot all three

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import f, t

n = 20
p = 2
nu = n - p - 1
rss = 576.2

sx1, sx1_2 = 1.8, 21.2
```

```

sx2, sx2_2 = 14.5, 83.7
sy = -66.6
sx1x2 = -4.0
sx1y = 34.6
sx2y = -127.1

x1bar = sx1 / n
x2bar = sx2 / n
ybar = sy / n

S11 = sx1_2 - n*x1bar**2
S22 = sx2_2 - n*x2bar**2
S12 = sx1x2 - n*x1bar*x2bar
S1y = sx1y - n*x1bar*ybar
S2y = sx2y - n*x2bar*ybar

G = np.array([[S11, S12], [S12, S22]], dtype=float)
Ginv = np.linalg.inv(G)

theta_hat = Ginv @ np.array([S1y, S2y])

alpha = 0.05
Fcrit = f.ppf(1-alpha, 2, nu)
c = 2*rss/nu * Fcrit

# Ellipse boundary in slope parameter coordinates
vals, vecs = np.linalg.eigh(G)
angles = np.linspace(0, 2*np.pi, 1000)
unit = np.vstack((np.cos(angles), np.sin(angles)))
ellipse = theta_hat[:, None] + vecs @ np.diag(1/np.sqrt(vals)) @ (np.sqrt(c)*unit)

# Bonferroni intervals
crit_bonf = t.ppf(1-alpha/(2*2), nu)
se = np.sqrt((rss/nu) * np.diag(Ginv))
bonf_half = crit_bonf * se

# Scheffe coordinate intervals
scheffe_half = np.sqrt(c * np.diag(Ginv))

print("theta_hat =", theta_hat)
print("Ginv =\n", Ginv)
print("ellipse half-widths =", scheffe_half)
print("Bonferroni half-widths =", bonf_half)

fig, ax = plt.subplots(figsize=(8, 7))
ax.plot(ellipse[0], ellipse[1], label="95% confidence ellipse")
ax.scatter(theta_hat[0], theta_hat[1], label="OLS estimate")

# Rectangle from ellipse projections
r1 = [theta_hat[0]-scheffe_half[0], theta_hat[0]+scheffe_half[0]]
r2 = [theta_hat[1]-scheffe_half[1], theta_hat[1]+scheffe_half[1]]
rect_x = [r1[0], r1[1], r1[1], r1[0], r1[0]]
rect_y = [r2[0], r2[0], r2[1], r2[1], r2[0]]
ax.plot(rect_x, rect_y, label="Smallest containing rectangle")

# Bonferroni rectangle
b1 = [theta_hat[0]-bonf_half[0], theta_hat[0]+bonf_half[0]]
b2 = [theta_hat[1]-bonf_half[1], theta_hat[1]+bonf_half[1]]
bonf_x = [b1[0], b1[1], b1[1], b1[0], b1[0]]
bonf_y = [b2[0], b2[0], b2[1], b2[1], b2[0]]

```

```

ax.plot(bonf_x, bonf_y, label="Bonferroni rectangle")

ax.set_xlabel(r"$\beta_1$")
ax.set_ylabel(r"$\beta_2$")
ax.set_title("95% simultaneous confidence regions for $(\beta_1, \beta_2)$")
ax.legend()
ax.grid(True, alpha=0.2)
plt.show()

```

Final conceptual summary

1. Centering makes the intercept orthogonal to the regressors, but it does *not* make the regressors orthogonal to one another.
2. Pairwise orthogonality makes multiple-regression slopes coincide with the corresponding separate simple-regression slopes.
3. Orthogonal regressors minimize the total coefficient variance when the regressor lengths are held fixed.
4. For every fixed linear functional $c^T \beta$, $c^T \hat{\beta}_{OLS}$ is BLUE.
5. With two regressors, multicollinearity enters the variance of $\hat{\beta}_1$ through the factor $1/(1 - r^2)$.
6. Under normality, $\hat{\beta}_{OLS}$ and RSS are independent; this is what allows exact t , F , ellipsoid, Bonferroni, and Scheffé procedures.
7. The MLE variance estimate RSS/n is biased but consistent; the unbiased estimator is $\text{RSS}/(n - p - 1)$.
8. LRTs for linear restrictions can be written either as likelihood ratios or equivalently as F statistics.
9. A simple correlation coefficient is linked to the LRT by $r^2 = 1 - \Lambda^{2/n}$.
10. Bonferroni protects finitely many specified contrasts by splitting the error probability; Scheffé protects an entire linear family and is therefore typically wider for any one pre-specified contrast.
11. The eigenvalues of $X^T X$ determine the precision of different parameter-space directions: large eigenvalue $\Rightarrow$ narrow confidence interval; small eigenvalue $\Rightarrow$ wide confidence interval.
12. Standardization preserves the fitted regression surface but changes the numerical coefficients, and it can change coefficient-equality restrictions because equality of slopes is scale-dependent.